Hadi Malik

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Education

Georgia Institute of Technology Atlanta, GA
Master of Science in Computer Science *May 2028*

Georgia Institute of Technology Atlanta, GA
Bachelor of Science in Computer Science | GPA: 4.0 *December 2026*
- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Systems & Networks, Deep Learning

Technical Skills

- **Languages**: Python, Java, C, C++, Rust, JavaScript, TypeScript, SQL, Go, Swift
- **Cloud & Tools**: AWS (Lambda, EC2, S3, DynamoDB, CloudWatch, CDK, Bedrock), Docker, CI/CD, Git, GCP, Azure
- **Web Development**: React, Node.js, REST APIs, API Gateway, PostgreSQL, MongoDB, MySQL, Django, Flask, Firebase
- **AI/MLOps**: PyTorch, TensorFlow, Scikit-learn, RAG, LangChain, OpenCV, Kubernetes, Keras, Slurm

Experience

Software Engineer Intern May 2026 - August 2026
Amazon *Seattle, WA*
- Built a **debugging and analytics platform** with React, S3, and CloudFront, used by **15+ Alexa Smart Home teams**.
- Cut **debug time by 4 hours** by building **data pipelines** with REST APIs and Lambda to merge DynamoDB/CloudWatch data.
- Enabled **sub-500ms analysis of millions of daily requests** by deploying **distributed systems** on Lambda in NA/EU/FE.
- Automated **99% of tests** by implementing **integration tests** with TestNG, JUnit, and seeded data to cover edge cases.

Machine Learning Research Assistant August 2025 - Present
Georgia Tech School of Cybersecurity and Privacy | Prof. Vassilis Zikas *Atlanta, GA*
- Co-authored the paper **"Causal Model Provenance via Shard Witness Auditing"** and submitted it to **AISTATS 2026**.
- Designed **fine-tuning tests** with Hugging Face Trainer, loss sequences, and MIN-K to identify training-data traces.
- Evaluated **KGW watermarking** with calibrated detectors to test whether training-data signals survive model fine-tuning.

Software Engineer Intern May 2025 - July 2025
In Good Company Works *Atlanta, GA*
- Created a **full-stack AI advisor** with React, Node.js, and Firebase, guiding **15 businesses across 50+ client sessions**.
- Reduced **onboarding time by 60%** by building **personalized RAG pipelines** with OpenAI Responses API and LangChain.
- Cut **backend latency by 50%** by creating **secure REST APIs** with Node.js on GCP, OAuth 2.0, and JWT authentication.
- Increased **user engagement by 20%** by shipping a companion **iOS app** with Swift for personalized mobile AI guidance.

Software Engineer Intern May 2024 - August 2024
Verifacto Inc. *Sandy Springs, GA*
- Achieved **99% accuracy** on insurance documents by fine-tuning a **LiLT classifier** with PyTorch/HF Transformers.
- Improved **query efficiency by 70%** by migrating **MySQL queries** to MongoDB with vector indexes for fast data retrieval.
- Built an **AI database chatbot** using Python/LangChain, cutting **time-to-insight by 60%** via natural-language queries.

Projects

Urology Report Generator | **Python, PyTorch, OpenCV, MedGemma, LoRA, Slurm** December 2025 - February 2026
- Attained **94.8% report similarity** with a **urology report generator** built using React/MedGemma, tuned on 600+ studies.
- Designed an **evaluation pipeline** with Python/BiomedCLIP to measure report similarity and flag potential hallucinations.
- Raised **image capacity 5x** and produced reports in **under 60 seconds** by optimizing **GPU memory** using FlashAttention.

Credit Card Fraud Detector | **Python, PyTorch, scikit-learn, PCA, DBSCAN** September 2025 - December 2025
- Built a **four-model fraud benchmark** with PyTorch/scikit-learn on 590K IEEE-CIS transactions, achieving **0.9007 ROC-AUC**.
- Developed a **PCA comparison pipeline** with scikit-learn, raising FraudNet ROC-AUC **from 0.8776 to 0.9007**.
- Created an **anomaly-detection benchmark** with K-Means/DBSCAN; DBSCAN reached **85.0% recall** and **82.3% accuracy**.